

A finite Hurwitz-kernel cocycle for periods and L -functions of meromorphic modular forms on $\mathrm{SL}_2(\mathbb{Z})$

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Abstract

We give one finite, regulator-free contour functional — a two-segment cocycle integral with an explicit Hurwitz-zeta T -kernel — that realizes both the period polynomials and the L -functions of modular forms on $\mathrm{SL}_2(\mathbb{Z})$ whose poles may lie anywhere: inside the fundamental domain and at the cusp. From a finite base point $\tau_0 \in \mathbb{H}$ one builds a cocycle whose cohomology class is independent of τ_0 . Removing its T -component by a coboundary — possible if and only if the constant Fourier coefficient $c_f(0)$ vanishes — yields the parabolic representative, whose S -value is the period polynomial. The same kernel, with the period monomial τ^ℓ replaced by τ^{s-1} , gives the completed L -function on all of $M_k^!$, constant term allowed, with the classical poles of Λ ; for forms with interior poles it is determined by the contour up to the residues crossed. The construction introduces no new objects: the periods are Brown’s quasi-periods in closed form [1], the L -values are those of Bringmann–Fricke–Kent [2]. The contribution is the explicit all- s kernel and the single finite realization. All formulae are checked numerically to 50 digits [12].

1 A finite L -integral

Definition 1.1 (S - and T -segments). *Let $\tau_0 \in \mathbb{H}$ and $f \in F_k$, and let $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ generate $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. The S - and T -segments at τ_0 are smooth paths in $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$,*

$$\gamma_{\tau_0}^S : -1/\tau_0 \longrightarrow \tau_0, \quad \gamma_{\tau_0}^T : \tau_0 - 1 \longrightarrow \tau_0, \quad (1)$$

running from $S^{-1}\tau_0 = -1/\tau_0$ and $T^{-1}\tau_0 = \tau_0 - 1$ to τ_0 .

The pole-avoidance is inert for $f \in M_k^!$: with no interior poles, $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty) = \mathbb{H}$ and the segments are unobstructed. Interior poles — and the homotopy classes of contours they create — become relevant only for meromorphic f , in the final section.

Definition 1.2 (Hurwitz T -kernel). *The Hurwitz T -kernel is, with $\zeta(s, x) = \sum_{m \geq 0} (m + x)^{-s}$ the Hurwitz zeta function (meromorphic in s , holomorphic in $x \in \mathbb{H}$),*

$$\tilde{k}_T(\tau, s) := \zeta(1 - s, \tau + 1) - e^{i\pi(s-1)} \zeta(s - (k-1), \tau + 1). \quad (2)$$

Definition 1.3 (L -integral). *Let $f \in F_k$, $\tau_0 \in \mathbb{H}$, and $s \in \mathbb{C}$, and let $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ be the S - and T -segments at τ_0 , as in Definition 1.1. The L -integral of f is*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := i^{-s} \left[\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right]. \quad (3)$$

Its τ_0 - and path-dependence is dealt with below, in the main body.

Definition 1.4 (*L-function*). *The L-function of $f \in F_k$ is*

$$L(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T). \quad (4)$$

Likewise; once that dependence is settled below, we drop the arguments and write simply $L^*(f, s)$ and $L(f, s)$.

Lemma 1.5 (Finiteness). *For $f \in F_k$, both integrals in Definition 1.3 are finite for every $s \in \mathbb{C}$, every $\tau_0 \in \mathbb{H}$, and every choice of paths $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$.*

Proposition 1.6 (Basepoint and path independence). *For $f \in M_k^!$ the value $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ is independent of $\tau_0 \in \mathbb{H}$ and of the paths: with no interior poles, $\mathbb{H}$ is simply connected, so all admissible contours are homotopic. We write $L^*(f, s)$ for the common value and $L(f, s) := \frac{(2\pi)^s}{\Gamma(s)} L^*(f, s)$.*

Remark 1.7 (Classical Mellin recovery). *For a holomorphic cusp form $f \in S_k$, the cusp limit $\tau_0 \rightarrow i\infty$ collapses the S -segment onto $[0, i\infty]$ while the T -segment vanishes exponentially, so*

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s) = \int_0^\infty t^{s-1} f(it) dt, \quad (5)$$

the classical completed L-integral.

2 The L-function

Proposition 2.1 (Cusp unfolding). *For $n > 0$ and $\nu \in \mathbb{C}$,*

$$\int_i^{i+1} e^{2\pi i n u} \zeta(\nu, u) du = i^{1-\nu} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n). \quad (6)$$

Theorem 2.2 (*L-function for weakly holomorphic forms*). *For $f \in M_k^!$, at $\tau_0 = i$,*

$$L^*(f, s) = -c_f(0) \left(\frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right] = i^k L^*(f, k-s). \quad (7)$$

Definition 2.3 (Period cocycle). *Let $V_n := \mathbb{C}[X, Y]_n$ with the action $(P|_\gamma)(X, Y) := P(aX + bY, cX + dY)$, and $\omega_f := f(\tau) (X - \tau Y)^n d\tau$. For $f \in F_k$ and $\tau_0 \in \mathbb{H}$, the period cocycle is*

$$C_\gamma^f(\tau_0) := (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} \omega_f \in V_n, \quad \gamma \in \Gamma, \quad (8)$$

satisfying $C_{gh}^f = C_g^f|_h + C_h^f$.

Lemma 2.4 (Parabolic representatives exist iff $c_f(0) = 0$). *With $\delta_T P := P|_T - P$ on V_n , one has $[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} c_f(0)$, so $C_T^f \in \text{im } \delta_T$ if and only if $c_f(0) = 0$ — the principal part of f never obstructs.*

Definition 2.5 (Period polynomial). *For $f \in S_k^!$ (so $c_f(0) = 0$), the period polynomial is the S -value of the parabolic representative,*

$$r_f := C_S^f(\tau_0) - (P|_S - P) \in V_n, \quad (9)$$

with $P \in V_n$ chosen so that $\delta_T P = C_T^f$ (Lemma 2.4).

Proposition 2.6 (Period coefficients are special L -values). *For $f \in S_k^!$ and $\ell \in \{1, \dots, k-3\}$,*

$$[r_f]_{X^{n-\ell}Y^\ell} = (2\pi i)^{n+1}(-1)^\ell \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1). \quad (10)$$

Corollary 2.7 (Period coefficients are quasi-period coordinates). *For every weight with $\dim S_k = 1$ — $k \in \{12, 16, 18, 20, 22, 26\}$ — and a weak cusp form $f \in S_k^!$, the coefficients of r_f (Proposition 2.6) are the quasi-period coordinates $\omega^\pm(f)$ in the sense of [1]: ω^+ from ℓ even, ω^- from ℓ odd. At $k = 12$ they reproduce Brown's published $\eta^\pm(\widehat{\Delta})$; at the remaining five weights they furnish the quasi-periods anew [12].*

Remark 2.8 (Period-determinant cross-check). *The extracted periods and quasi-periods satisfy, numerically, a Legendre-type relation: the determinant $\omega^+\eta^- - \omega^-\eta^+$ lies in $\mathbb{Q}^\times \cdot (2\pi i)^{k-1}$, equal to $-(k-2)!(2\pi i)^{k-1}$ at $k = 12$ in Brown's normalization (the weight- k analog of $\omega_1\eta_2 - \omega_2\eta_1 = 2\pi i$). This is a global consistency check on the extraction, claiming nothing new [12].*

2.1 Why a finite contour

Remark 2.9. *For a weak cusp form $f \in S_k^!$ the classical Mellin integral $\int_0^{i\infty} f(\tau)(\tau/i)^s \frac{d\tau}{\tau}$ does not converge — f grows at the cusp — and the deformed-Mellin construction of [2] supplies a finite value through a non-holomorphic regularization there. The functional of Definition 1.3 reaches the same value with no regularization: written as its two segments, each diverges as $\tau_0 \rightarrow i\infty$, with opposite signs, and the two cancel at every finite τ_0 , so $L^*(f, s)$ is finite throughout $\mathbb{H}$ although the Mellin contour itself has no limit. By Proposition 1.6 the value is independent of τ_0 and of the paths; BFK's regularized value is its evaluation at $\tau_0 = i$ (Theorem 2.2), and its integer- s values are Brown's quasi-periods (Corollary 2.7). Two independent finite computations — Brown's cohomological periods and the BFK L -values, obtained without reference to one another — meet at this one contour, which fixes it as the canonical object. Finiteness and τ_0 -independence are the analytic shadow of the parabolic representative [1, 5, 6].*

3 Meromorphic modular forms

Definition 3.1 (Polylog projection). *Let $f \in F_k$ have interior poles at the points τ_p of a set $\mathcal{P}_f$ of Γ -inequivalent representatives, and write $\text{Li}_s(z) = \sum_{n>0} z^n n^{-s}$. Since $\text{Li}_{1-m}(e^{2\pi i(\tau-\tau_p)})$ has a pole of order m at τ_p , the principal part of f there is matched by a unique polylog projection*

$$\widehat{P}_{\tau_p} f(\tau) := \sum_{m>0} r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e(\tau - \tau_p)), \quad (11)$$

its coefficients $r_{f, \tau_p}^(m)$ being the principal-part data of f at τ_p . The projection $\widehat{P} := \sum_{\text{im } \tau_p \geq 1} \widehat{P}_{\tau_p}$ removes the principal part of every pole on or above the line $\text{im } \tau = 1$.*

Definition 3.2 (Regularized cusp coefficients). *With $\widehat{P}$ as in Definition 3.1,*

$$c_{Rf}(n) := [(I - \widehat{P})f]_{q^n}. \quad (12)$$

Definition 3.3 (The space $F(\infty)$). *$F(\infty)$ is the space of meromorphic g on $\mathbb{H}$ with $g(\tau) = g(\tau+1)$, finitely many poles in any compact subset of the strip $\{\text{Im } \tau > 0, |\text{Re } \tau| \leq \frac{1}{2}\}$, and $g(\tau) = O(e^{C \text{Im } \tau})$ as $\text{Im } \tau \rightarrow \infty$ for some C . Each such g has a Fourier expansion $g = \sum_n c_g(n) q^n$.*

Then $F_k \subset F(\infty)$, and $(I - \widehat{P})f \in F(\infty)$ — the polylog tails are T -periodic but not modular, so the projection leaves F_k .

Lemma 3.4 (*q-series–integral commutation*). *Let $g \in F(\infty)$ have no pole on or above the line $\text{Im } \tau = 1$ (e.g. $g = (I - \widehat{P})f$). Then $g = \sum_n c_g(n)q^n$ converges along the T -segment at $\tau_0 = i$ and integrates term by term:*

$$\int_{\gamma_i^T} g(\tau) \widetilde{k}_T(\tau, s) d\tau = \sum_n c_g(n) \int_{i-1}^i e^{2\pi i n \tau} \widetilde{k}_T(\tau, s) d\tau. \quad (13)$$

A pole at $\text{Im } \tau_p \geq 1$ defeats this — there the coefficients grow at least like $n^{m-1}e^{2\pi n \text{Im } \tau_p}$ [3, Lem. 2.6].

Definition 3.5 (Polylog integral). *For $N \geq 0$ and $z \in \mathbb{H}$, the polylog integral is the L -integral of a single tail $\text{Li}_{-N}(e(\tau - z))$ along the T -segment at $\tau_0 = i$ (where γ^S degenerates, since $-1/i = i$):*

$$I_N(s, z) := i^{-s} \int_{i-1}^i \text{Li}_{-N}(e(\tau - z)) \widetilde{k}_T(\tau, s) d\tau. \quad (14)$$

Lemma 3.6 (Closed form of the polylog integral). *For $\text{Im } z < 1$, the polylog integral is the cusp unfolding (Proposition 2.1) summed against the tail’s coefficients:*

$$I_N(s, z) = \sum_{n \geq 1} n^N e^{-2\pi i n z} \left[\frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k - s, 2\pi n)}{(2\pi n)^{k-s}} \right]. \quad (15)$$

For $\text{Im } z \geq 1$ — the projected poles — $I_N(s, z)$ is the analytic continuation in z of the right-hand side, corrected by the residue crossed at $\text{Im } \tau = 1$ (Lemma ??).

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